

Fig. 4: Basic unit of thermoelectric generator

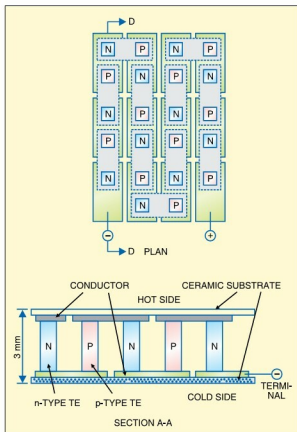


Fig. 5: Thermoelectric module for power generation

thermoelectric material is normally determined by lattice conduction. The basic unit of a thermoelectric converter consists of two types of thermoelectric semiconductors, and it is desirable to have a base material that can be both p- and n-type doped, so that the same material system can be used on both sides of the junctions.

Figure of merit ' Z ' expresses the efficiency of the thermoelectric couple:

$$Z = S^2 \alpha / \lambda$$

where ' S ' is Seebeck coefficient, ' α ' is electrical conductivity and ' λ ' is thermal conductivity.

In order to maximise the figure of merit, the best semiconductor material is to be chosen with controlled doping. The smaller the density of electrons or holes, the larger the Seebeck coefficient. However, if the density of charge carriers becomes too low, the electrical resistivity will rise so high that the figure of merit is reduced. ' Z ' reaches maximum value when the density of electrons or holes becomes one per 10,000 atoms. At this juncture, the Seebeck coefficient is about 200 microvolts per degree celsius.

Lattice thermal conductivity can vary by a factor of 1000 between different semiconductors. It is the most important variable in selecting an optimal thermoelectric material. Materials made of heavy atoms like lead and bismuth have low lattice thermal conductivity. Hence compound semiconductors like bismuth telluride and lead telluride are useful thermoelectric materials.

Another factor that reduces thermal conductivity is induction of disorderliness of the atomic arrangement in crystals. In bismuth telluride if some of the atoms of bismuth are replaced with antimony or some of tellurium

is replaced with selenium, it would reduce the thermal conductivity by a factor of about 10 while retaining the same electrical properties. This would increase ' Z ' by a factor of more than 2.

In today's best commercial thermoelectric modules, ' Z ' is about 1.0. To maximise power-generation efficiency, ' Z ' and the temperature differential between hot and cold sides should be as large as possible.

Nanotechnology and its quantum-scale synthesis lead to a new super thermoelectric material that gives higher ' Z ' by confining electrons to two-di-

mensional quantum wells. At the same time, thermal conductivity is suppressed. Recently, researchers working on this concept have developed super thermoelectric materials that can achieve ' Z ' of about 2.4 for a nanoscale structure made by alternating layers of two thermoelectric materials that both enhance the Seebeck coefficient and suppress thermal conductivity at room temperature with conversion efficiency of about 20 per cent.

The basic unit of a thermoelectric generator is shown in Fig. 4. It consists of two types of high-performance thermoelectric elements like bismuth telluride or silicon germanium. One is p-type and the other is n-type semiconductor. At one end, both the elements are connected together with a metal of high conductivity, enabling electrons to flow from n-type to p-type leg and also uniform hot-side temperature. On cold side, two metallic electrodes are connected to the legs, separately forming positive and negative power terminals. The thermoelectric is electrically separated from the hot and cold fluids by a suitable insulator, so there is no short circuit or leakage of electricity.

Due to temperature differential, electron-hole pairs are created at the hot end, absorbing heat in the process. The pairs recombine and reject heat at the cold ends. A voltage potential—the Seebeck voltage which drives the hole/electron flow—is created between the hot and cold ends of the thermoelectric elements. The net voltage appears across the bottom of the thermoelectric element legs. Electrical connections can be made from the thermoelectric elements to an external load to extract power.

Our body loses a large amount of energy as heat at the rate of about 100 watts, even when we are just sitting and listening to music. Some of that heat energy, precisely the temperature difference between the skin and the air, can be used to make electrons flow, thus generating electricity. This thermoelectric effect has been known